

Kathryn E. Windsor

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Education

PhD Candidate in Cell and Molecular Biology
The University of Texas at Austin

May 2028 (Expected)

B.S. in Neuroscience and Spanish, Minor in Religion
Trinity University, San Antonio, Texas
GPA: 3.52, *Cum Laude*

May 2019

Research Publications

- Ghatpande, V., Lewis, A. J., **Windsor, K. E.**, Lee, H., Alcalá, A. J., Menke, D. B., Cenik, C. & Vokes, S. A. (2025). Developmental and transcriptional programs that define the initiation of the forelimb. *Development* (In Review).
- Ramachandran, J., Chen, W., Lex, R. K., **Windsor, K. E.**, Lee, H., Wang, T., Zhou, W., Ji, H., & Vokes, S. A. (2023). The BAF chromatin complex component SMARCC1 does not mediate GLI transcriptional repression of Hedgehog target genes in limb buds. *Developmental Biology*, 504, 128–136.
- Lex, R. K., Zhou, W., Ji, Z., Falkenstein, K. N., Schuler, K. E., **Windsor, K. E.**, Kim, J.D., Ji, H., & Vokes, S. A. (2022). GLI transcriptional repression is inert prior to Hedgehog pathway activation. *Nature communications*, 13(1), 1-15.
- Moreno Manrique, J. F., Voit, P. R., **Windsor, K. E.**, Karla, A. R., Rodriguez, S. R., & Beaudoin III, G. M. (2021). SynapseJ: an automated, synapse identification macro for ImageJ. *Frontiers in Neural Circuits*, 97.

Research Presentations

- **Windsor, K.**, Jursa, T., Smith, D. R., Mukhopadhyay, S. (2026, March 22-26). *Role of brain HIF signaling in motor function and manganese neurotoxicity*. [Poster Presentation]. Society of Toxicology, San Diego, California, United States.
- **Windsor, K.** (2025, May 5). *How Your Brain Grows and Learns* [Elementary school presentation]. Lee Elementary Science Monday, Austin, Texas, United States.
- **Windsor, K.** (2025, April 25). *Manganese and Your Brain* [All ages presentation]. Austin Homeschoolers Science Club, Austin, Texas, United States.
- **Windsor, K.** (2025, February 21; 2024, November 7). *Manganese and Your Brain* [Middle school presentation]. Kealing Middle School, Austin, Texas, United States.
- **Windsor, K.** (2024, September 13-15). *Dissecting Mechanisms of Neuronal Manganese Homeostasis* [Department presentation]. Interdisciplinary Life Science Graduate Program Annual Retreat, Austin, TX, United States.
- **Windsor, K.**, Voit, P., Litch, A., Hemani, S., Karla, A., Guo, C., Ferrer, P., & Beaudoin III., G. M. (2018, November 3-7). *Structural and functional changes of input-specific excitatory synapses on nigral dopamine cells caused by cocaine*. [Poster Presentation]. Society for Neuroscience, San Diego, California, United States.

Other Publications & Media

- **Windsor, K.** (2026, March). Hungarian Mummies of the White Plague: Homeotic Transformation in a Historic Epidemic. *Society for Developmental Biology*. (In Review).
- **Windsor, K.** (2025, December 23). How do electric eels work? New observations on the world's oldest (organic) battery. *Society for Developmental Biology*. <https://doi.org/10.1016/j.ydbio.2025.12.019>

- **Windsor, K.** (2025, November 21). A Dash for the finish line, with help from GetHIRED! *Society for Developmental Biology*. <https://www.sdbonline.org/resource?ResourceID=4534>
- **Windsor, K.** (2025, August 11). UT science communication is by students, for students. *The Daily Texan*. <https://thedailytexan.com/2025/08/11/ut-science-communication-is-by-students-for-students/>
- **Windsor, K.** (2025, August 8). Autism research should not be at the mercy of RFK Jr.'s ignorance. *The Daily Texan*. <https://thedailytexan.com/2025/08/08/autism-research-should-not-be-at-the-mercy-of-rfk-jr-s-ignorance/>
- **Windsor, K.** (2025, July 21). Revolution is driven by art. *The Daily Texan*. <https://thedailytexan.com/2025/07/21/revolution-is-driven-by-art/>
- **Windsor, K.** (2025, July 13). Cognitive offloading: why do more, when you could do less? *The Daily Texan*. <https://thedailytexan.com/2025/07/13/cognitive-offloading-why-do-more-when-you-could-do-less/>
- **Windsor, K.** (2025, July 7). Academia isn't dead, but it's losing ground. *The Daily Texan*. <https://thedailytexan.com/2025/07/07/academia-isnt-dead-but-its-losing-ground/>
- **Windsor, K.** (Host). (2024). *She Blinded Me with Science!* [Audio podcast]. KVRX. <https://shows.acast.com/sciencekvrX>

Research Experience

Graduate Research Assistant

January 2024 – Present

University of Texas, Austin, TX

Supervisor: Dr. Somshuvra Mukhopadhyay

- Investigate the role of hypoxia inducible factor isoforms in manganese-induced parkinsonian syndrome
- Manage multiple mouse lines, including daily treatments and behavioral testing

Research Assistant

August 2020 – June 2023

University of Texas, Austin, TX

Supervisor: Dr. Steven Vokes

- Conducted primary research in embryonic limb buds to elucidate mechanisms of the *Hedgehog* signaling pathway
- Maintained mouse colony and breed new transgenic lines
- Managed lab inventory and ordering to support ongoing experiments

Research Assistant

June 2019 – July 2020

Austin Speech Labs, Austin, TX

Supervisor: Shilpa Shamapant

- Designed and implemented experimental therapy conditions to increase speech recovery in stroke survivors with Broca's aphasia
- Administered bilingual speech therapy to measure effect of therapy language on anomia rehabilitation
- Recorded speech ability assessment scores and created visual representations of clients' progress

Undergraduate Research Assistant

August 2017 – May 2019

Trinity University, San Antonio, TX

Department of Biology

Advisor: Dr. Gerard Beaudoin III

- Conducted primary research on structural changes of input-specific excitatory synapses on nigral dopamine cells caused by cocaine
- Developed ImageJ macro to analyze colocalization and intensity of signal

- Mentored undergraduate students in lab techniques

Undergraduate Research Assistant

August 2016 – May 2017

Trinity University, San Antonio, TX
Department of Biology
Advisor: Dr. James Roberts

- Maintained cell lines to investigate role of estrogens in mediating neural protection and recovery from damage due to oxidative stress

Undergraduate Research Assistant

August 2016 – December 2016

Trinity University, San Antonio, TX
Department of Psychology
Advisor: Dr. William Ellison

- Recruited and coordinated undergraduate research participants
- Encoded participant data to model heterogeneity and stability of dynamic factors in within-person variability of personality

Awards

- **Dr. Marsha Wheeler Miller Fellowship Recipient** **May 2024**
- **Alamo Brain Bowl Grand Champion**, Trinity University Brain Bowl Team **Spring 2019**
- **Trainee Professional Development Award**, Society for Neuroscience **September 2018**

Service & Outreach

- **President** **May 2025 - Present**
Present your PhD to a 12-year-old, University of Texas at Austin
- **Judge** **September 2025**
Fall Undergraduate Research Symposium Science Slam, University of Texas at Austin
- **Volunteer Pen Pal** **September 2024 - Present**
Letters to a Pre-Scientist (National STEM Outreach Program)
- **Science Fair Judge**, 4th – 7th Grade **March 2019**
St. George Episcopal School, San Antonio, TX
- **President, Treasurer** **April 2016 – December 2018**
Neuroscience Student Association, Trinity University

Teaching Experience

- **Teaching Assistant**, Integrative Biology Lab **Spring 2019**
Department of Biology, Trinity University, San Antonio, TX
- **Teaching Assistant**, 1st – 8th Grade Science **Summer 2017**
Department of Science, Real Colegio Nuestra Señora del Loreto, Madrid, Spain

Skills

- Molecular Biology Techniques: Immunofluorescence, Light & Confocal microscopy, Cryostat sectioning, PCR genotyping, RNA isolation, qPCR, Skeletal preparation, Western blot, Cell culture, Molecular cloning
- Animal techniques: Mouse handling, Breeding, Colony management, Behavioral assays, Euthanasia, Dissection, Intraperitoneal injection, Perfusion
- Data analysis: Microsoft Excel, SPSS, ImageJ, R, Prism
- Fluent in Spanish